MIL-D-18616(NOrd)

13 June 1955

SUPERSEDING
NAVORD OS 5961D
9 FEBRUARY 1955

MILITARY SPECIFICATION

DETONATOR MARK 56 MOD 0

1. SCOPE

1.1 This specification governs the manufacture, assembly, loading, and preparation for delivery of Detonator Mk 56 Mod 0 and the methods of inspection and tests upon which acceptance will be based. Detonator Mk 56 Mod 0 is stab initiated.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawings, and publications, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

MILITARY

JAN-T-339 -- Tetryl (Trinitrophenylmethylnitramine).

STANDARDS

MIL-STD-105 -- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 -- Marking for Shipment and Storage.

DRAWINGS

BUREAU OF ORDNANCE

LD 275955 -- Detonator Mk 56 Mod 0,
LD 166538 -- Test Set (Detonator) Mk 136 Mod 0,

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LD 275959 -- Detonator Container Mk 6 Mod 0 (Inner),
LD 275960 -- Detonator Container Mk 8 Mod 0 (Outer),
LD 284563 -- Test Set Mk 170 Mod 1,
398475 -- Firing Pin and Detonator Holder,
553333 -- Anvil,
553491 -- Test Ball,
1384952 -- Test Sleeve,
and all documents listed on the above LDs and drawings.

PUBLICATIONS

BUREAU OF ORDNANCE

OP 400 -- General Instructions for the Design, Manufacture, and Inspection of Naval Ordnance Equipment.
OD 5823 -- Sensitivity Test of Primers and Detonators using Test Set Mk 135 Mod 0 (Primers) and Test Set Mk 136 Mod 0 (Detonators).

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the Procuring Activity or as directed by the Contracting Officer.)

2.2 Other publications. -- The following documents form a part of this specification. Unless otherwise indicated the issue in effect on date of invitation for bids shall apply.

CODE OF FEDERAL REGULATIONS

49 CFR 71-78 -- Transportation, Interstate Commerce Commission, Explosives and Other Dangerous Articles.

(The Interstate Commerce Commission regulations are now a part of the Code of Federal Regulations (1949 Edition-Revised 1950) available from the Superintendent of Documents, Government Printing Office, Washington 25, D.C. Orders for the above publication should cite "49 CFR 71-78 (Rev. 1950)".)

3. REQUIREMENTS

3.1 General. -- Detonators Mk 56 Mod 0 covered by this specification shall be manufactured in accordance with Bureau of Ordnance List of Drawings LD No. 275955 and all documents listed thereon.

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3.1.1 The requirements and procedures established in OP 400 shall be adhered to wherever applicable.

3.2 Priming mixture. -- The moisture content of the priming mixture at the time of loading shall be less than 0.30 percent as determined by the procedure specified in 4.5.1. The composition of the priming mixture shall meet the drawing requirements when subjected to the chemical analysis specified in 4.5.3.

3.3 Lead azide. -- The moisture content of the lead azide at the time of loading shall be less than 0.30 percent as determined by the procedure specified in 4.5.1.

3.4 Tetryl. -- The volatiles and moisture content of the tetryl at the time of loading shall not exceed 0.10 percent when tested as specified in 4.5.2.

3.5 Detonator loading. -- The charge weights of the explosive components of the detonator shall be as specified on the applicable drawing. The conformance of these weights to the drawing requirements shall be controlled and determined by the procedures specified in 4.5.4 and 4.5.5.

3.6 Detonator sensitivity. -- The sensitivity of the detonator shall be ascertained as specified in 4.5.6. The sensitivity value shall be used for data card record and process control. Deviations in sensitivity values shall not directly cause the rejection of detonators, but may be reason for checking materials, processes, and priming mixture composition for compliance with drawing and specification requirements.

3.7 Detonator functioning. -- The detonator shall function on firing pin penetration when tested in Test Set (Detonator) Mk 136 Mod 0 using a two ounce test ball and a drop height of 12 inches. Testing and acceptance shall be as specified in 4.5.7.

3.8 Detonator output. -- Explosion of the detonators in Test Set Mk 136 Mod 0 shall produce perforations in the lead discs for 85 percent of the detonators tested. The average diameter of the perforations shall be 0.200 inch or greater. The perforation diameter shall be the diameter of a circle of equal area as determined by the use of Test Set Mk 170 Mod 1. Each individual indentation produced in lead discs which are not perforated shall be 0.195 inch or greater in depth. Testing and acceptance shall be as specified in 4.5.8.

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4. QUALITY ASSURANCE PROVISIONS AND TEST REQUIREMENTS

4.1 Sampling. -- Unless otherwise specified, and when applicable, the sampling plans and procedures used by the Government Inspector in the determination of the acceptability of products submitted by a supplier for Government Inspection will be in accordance with the provisions of Standard MIL-STD-105.

4.1.1 Lot. -- As applied to the Government Inspection of units of product, the term "Lot" shall mean "Inspection Lot", i.e., a collection of units of the product used as a basis for Government Inspection. Unless otherwise specified, the number of units of product in "Inspection Lots" shall be as determined by the Government Inspector, and may differ from the quantity designated in the contract or order as a lot for production, shipment, or other purpose.

4.2 Preproduction sample. -- Before entering into quantity production, a preproduction sample of one thousand (1000) Detonators Mk 56 Mod 0 shall be delivered by the Contractor for test to an agency designated by the Procuring Agency. These detonators shall be manufactured by the same procedures and processes and at the same location proposed by the Contractor for the execution of the contract. When a preproduction sample is received which proves to be satisfactory for the purpose intended, the Contractor will be notified and will be authorized by the Procuring Agency to proceed with the submission of the product for acceptance. Accepted samples will become the property of the Procuring Agency and will be included in the number called for in the contract schedule.

4.2.1 Preproduction sample for a subsequent contract. -- The necessity for a preproduction sample will be determined by the Procuring Agency when production under a new contract by the same Contractor at the same location follows the manufacture of satisfactory material as covered by this specification.

4.3 Production lot. -- For procurement purposes each lot of loaded detonators shall consist of 10,000 units plus those required for all test purposes. Unless otherwise stated in the contract, order, or requisition all units for test purposes shall be at the expense of the Contractor. The production lot should not be confused with the lot for inspection purposes as defined in 4.1.1.

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4.4 Inspection. -- The general inspection procedures shall be in accordance with "General Instructions for the Design, Manufacture, and Inspection of Naval Ordnance Equipment", OP 400; Government Inspection shall be in accordance with the provisions of General Procedures for Government Inspection and Contractor's Inspection shall be in accordance with Inspection by Contractor, as contained in OP 400.

4.5 Tests.

4.5.1 Moisture content test for priming mixture and lead azide. Conduct at least one test for each lot of detonators and such additional tests as the Inspector considers warranted by variations in the process and materials. Collect an approximate two gram sample of the material being loaded in a tared, stoppered container at the site of and immediately prior to consolidating the charge in the detonator. Weigh the container; open and expose the sample for drying at 50 ± 2 degrees C and at least 28 inches (Hg) vacuum until constant weight is reachod. Cool the container in a desiccator; stopper and weigh. All weighings shall be accurate to ± 0.5 milligram. The presence of moisture in excess of 0.30 percent for the priming mixture or the lead azide shall cause suspension of the charge loading until permanent corrective measures satisfactory to the Inspector have been completed. Detonators which have been loaded with defective charges shall be rejected.

4.5.2 Volatiles and moisture content test for tetryl. -- Conduct at least one test for each lot of detonators and such additional tests as the Inspector considers warranted by variations in the process and materials. Test procedure shall be in accordance with that given in Specification JAN-T-339. The presence of volatiles and moisture in excess of 0.10 percent shall cause suspension of the charge loading until permanent corrective measures satisfactory to the Inspector have been completed. Detonators which have been loaded with defective charges shall be rejected.

4.5.3 Chemical analysis of priming mixture. -- Chemical analysis of the priming mixture to determine conformity with composition requirements of the drawing shall be made at the request of the Inspector. The test will normally be employed to determine cause of marked deviations in sensitivity or functioning resulting from composition or workmanship deficiencies. Priming mixture batches and detonators loaded therefrom shall be rejected when the composition does not meet drawing requirements. Analytical procedures

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for determining the percentage of each constituent of the priming mixture shall be in accordance with 4.5.3.1 through 4.5.3.4.

4.5.3.1 Potassium chlorate. -- Weigh accurately approximately one gram of the priming mixture which has been dried in accordance with 4.5.1. Transfer the sample to a tared ignited filtering crucible of approximately 15 ml capacity. Add three ml of a saturated solution of lead azide in distilled water prepared at a temperature of 25 ± 2 degrees C. Agitate for 60 seconds. If necessary, break up any lumps with a glass rod taking care not to disturb the filter mat. Apply suction. Repeat this procedure five times, making a total of six 3 ml extractions. Rinse the sample in the filtering crucible with a few ml of alcohol, then ether; dry for approximately 15 minutes at 95 degrees C; cool in a desiccator and weigh. The percentage loss of weight, based on the weight of the original sample, shall be considered the percentage of potassium chlorate.

4.5.3.2 Lead azide. -- Extract the residue from the potassium chlorate determination with 5 ml portions of saturated ammonium acetate solution at about 25 degrees C. Hot ammonium acetate solution must not be used as it dissolves the antimony sulphide to some extent. Agitate each extraction for about 30 seconds and repeat extractions until the filtrate no longer gives a yellow precipitate with potassium dichromate solution. The use of more than 60 ml of ammonium acetate should be avoided. Wash the residue with distilled water; rinse with a few ml of alcohol, then ether; dry for approximately 15 minutes at 100 degrees C; cool in a desiccator, and weigh. The percentage loss of weight, based on the weight of the original sample, shall be considered the percentage of lead azide.

4.5.3.3 Antimony sulphide. -- Extract the residue from the lead azide determination with hydrochloric acid until free from antimony sulphide. Cold concentrated hydrochloric acid is used for the extraction. After neutralizing, ammonium sulphide may be used to detect antimony in the filtrate. Wash the crucible contents with hot concentrated hydrochloric acid. Ignite until free from sulphur, cool in a desiccator, and weigh. The percentage loss of weight, based on the weight of the original sample, shall be considered the percentage of antimony sulphide.

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4.5.3.4 Carborundum. -- Subtract the original weight of the ignited filtering crucible from the weight of the crucible and residue after the antimony sulphide determination. This difference, calculated as a percentage of the original sample, shall be considered the percentage of carborundum.

4.5.4 Process control tests for detonator charge weights. -- Process control tests shall be conducted for each variation in the operating procedures, processes, or materials and at such other times as the Inspector considers warranted.

4.5.4.1 Process control procedure. -- Make five inspections on individual detonators for the conformance of each increment charge weight with the drawing requirements. A deviation from drawing requirements shall cause a suspension of that phase of the explosive loading until permanent corrective measures satisfactory to the Inspector have been completed.

4.5.5 Increment charge weight test. -- From each maximum lot of ten-thousand (10,000) detonators select at random times during manufacture ten detonators at each of the three incremental stages of loading. This selection will provide 30 partially and fully charged detonators for ten individual measurements of each increment charge weight. Each measurement shall be to the nearest milligram. The lot shall be rejected if the average of a sample of ten measurements on an increment charge weight plus or minus 1.5 standard deviations of the values exceeds the minimum or maximum weight specified for the increment on the drawings. Detonators from the process control tests shall not be used to fill the sample requirements for this test.

4.5.5.1 Computation of the average and standard deviation of a sample. -- NAVORD Form 1844 (New 8-47) describes the calculation procedure for the computation of the average and standard deviation of a sample and provides a convenient form for recording of the data.

4.5.6 Detonator sensitivity test. -- Select 50 detonators from each maximum lot of 10,000 detonators for the Bruceton (Staircase) Sensitivity Test employing 0.5 inch increments. Use Test Set (Detonator) Mk 136 Mod 0, Bureau of Ordnance List of Drawings LD No. 166538, with Test Ball, Test Sleeve, Anvil, and Firing Pin and Detonator Holder Bureau of Ordnance Drawings 553491 piece 3, 1384952, 553333 piece 1, and 398475 piece 9 respectively. Test set operation and test

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procedures are described in OD 5823. Record, on the Ammunition Data Card specified in 5.4, the 50 percent firing height ($\bar{h}$) and the standard deviation (s) calculated from the test data.

4.5.7 Detonator functioning test. -- Samples shall be selected and tested from each lot in accordance with the Multiple Sampling Plan, Inspection Level II, AQL 0.25 percent defective as given in Standard MIL-STD-105. The initiation of the detonator shall be accomplished by the use of Test Set Mk 136 Mod 0 equipped as specified in 4.5.6. The two ounce test ball shall be dropped from a height of 12 inches to initiate the detonator. Detonators which do not explode on the first penetration of the firing pin shall be classed as defective. Acceptance of the lot shall be in accordance with the sampling plan given above. All the perforated or indented lead discs from the test shall be retained and used for filling the sample requirements of 4.5.8.

4.5.8 Detonator output test. -- On lots of 22,000 detonators or less, one hundred and fifty (150) lead discs shall be selected from the discs obtained from tests of 4.5.7. The order of selection shall be the firing order. In small lots the sample deficiency shall be supplied by extra firings to provide a sample of at least 150 specimens. On lots greater than 22,000 detonators, all the perforated or indented lead discs from tests of 4.5.7 shall be used for ascertaining output. Acceptance of the lot shall be based on the following criteria:

(a) The detonators tested shall produce holes in 85 percent or more of the lead discs. Failure to meet this requirement shall cause rejection of the lot from which the samples were taken.

(b) The average of the diameter of the perforation in all the perforated lead discs shall be equal to or greater than 0.200 inch as determined by measurement with Test Set Mk 170 Mod 1. Failure to meet this requirement shall cause rejection of the lot from which the samples were taken.

(c) The depth of each lead disc indentation produced by the explosion of the detonators shall be equal to or greater than 0.195 inch when measured as specified in 4.5.8.1. Failure of two percent or more of the total

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sample of output test detonators to meet this requirement shall cause rejection of the lot from which the sample was taken.

4.5.8.1 Measurement of lead disc indentations. -- Prior to measurement of the depth of indentations, brass or aluminum particles imbedded in the lead disc which might interfere with the measurement shall be removed by the use of a sharp pointed instrument. Care shall be taken to prevent gouging of the lead disc. Mount a dial indicator or comparator vertically on a stand so that the tip of the indicator shaft is perpendicular to the surface plate. The indicator shaft shall be 0.100 inch in diameter and have a 60 degree conical tip with 0.025 inch radius on the point. Place the lead disc indented side up, on a ring type holder so that the periphery of the bottom of the lead disc is parallel to the surface plate. Lower the indicator tip to the bottom of the indentation of the lead disc and set the dial reading at zero. The lead disc measurement shall be the average of the readings of four equidistant points near the periphery of the upper surface of the lead disc.

5. PREPARATION FOR DELIVERY

5.1 Packaging. -- One-thousand (1000) Detonators Mk 56 Mod 0 shall be packaged in accordance with Bureau of Ordnance List of Drawings LD No. 275959.

5.2 Packing. -- Four unit containers as specified in 5.1, shall be packed in accordance with Bureau of Ordnance List of Drawings LD No. 275960.

5.3 Marking. -- In addition to any special marking required by the contract or order, all containers shall be marked in accordance with Standard MIL-STD-129 and Code of Federal Regulations 49 CFR 71-78.

5.4 Data cards. -- Data cards, five by eight inches, printed as shown below, shall be filled out for each lot of detonators. Two of these cards shall accompany the shipping ticket; three shall be forwarded to the Procuring Agency, and five shall be furnished the Inspector.

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AMMUNITION DATA CARD

Mark No. _____ Spec. No. _____

Dwg. No. _____ Date (Mo. & Yr.) _____

Loaded by _____ Req. No. _____

Contract No. _____ No. in Lot _____

No. in Box _____ $\bar{h}$ = _____

Lot No. _____ s = _____

MATERIAL	MANUFACTURER	LOT NO.	DATE (MO. & YR.)
CUP			
CLOSING DISC			
LEAD AZIDE			
TETRYL			
POTASSIUM CHLORATE			
ANTIMONY SULPHIDE			
CARBORUNDUM			
PRIMING MIXTURE			

Remarks: (These are to include any changes in materials or manufacturing processes prescribed by this specification which the Inspector feels might affect the uniformity of the lot.)

6. NOTES. -- There are none applicable.

Patent notice. -- When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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